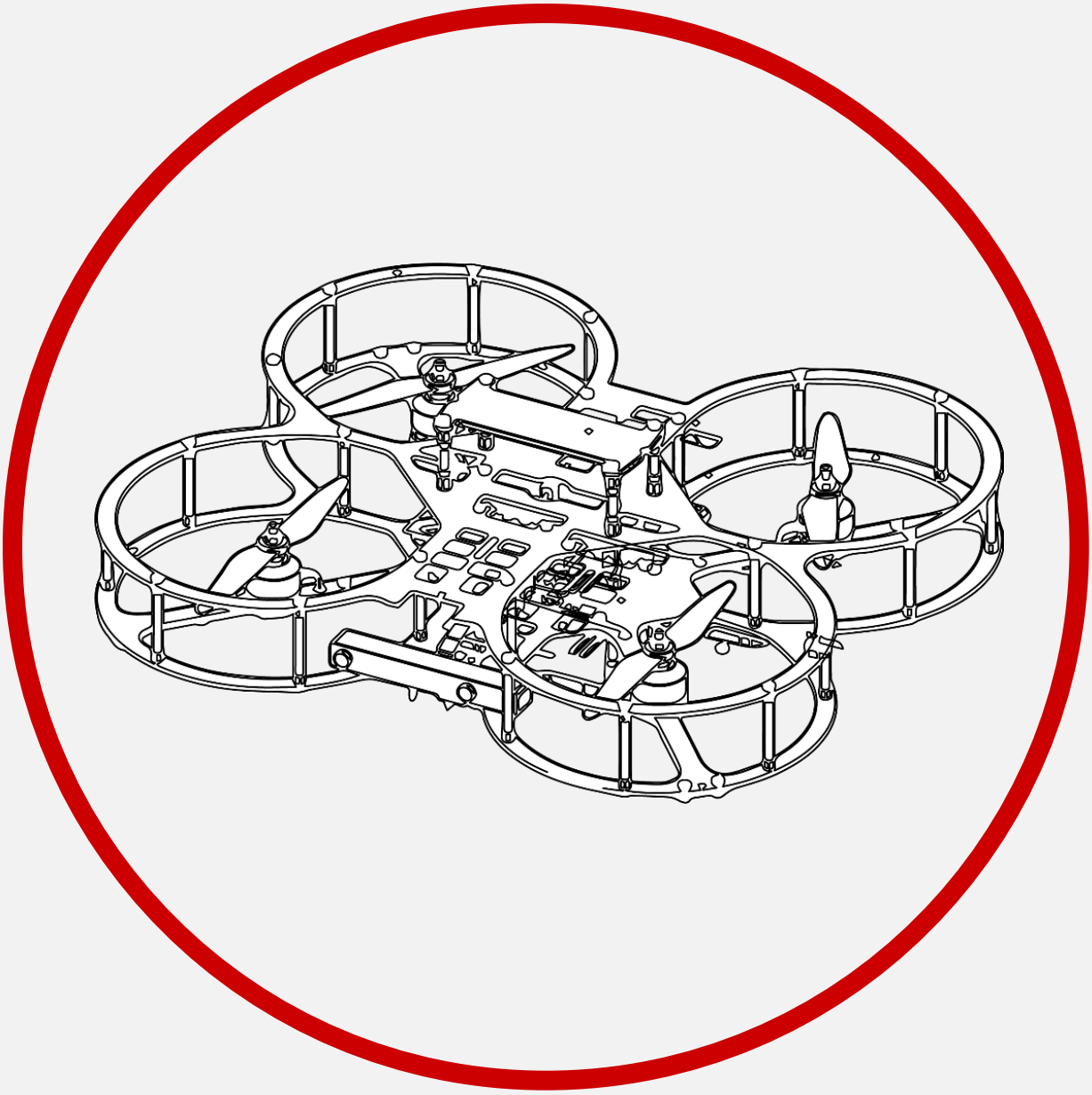

Autonomous Vehicles Research Studio

Setup Guide – QDrone 1 Batteries

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A. LiPo Battery Safety

Caution: Before using any batteries, chargers/balancers, or power supplies, users must first read the manuals packaged with their equipment. Quanser supplies these guidelines for charging batteries, but it is the users' responsibility to ensure they are operating their equipment safely and correctly. Quanser is not responsible for any damages resulting from use of batteries, power supplies, chargers, or balancers.

Caution: Prior to using the QDrone, visually check the battery for bloating or damage. If the battery exhibits bloating **DO NOT USE** it. Visual bloating of the battery is dangerous - discard it in accordance with your country's relevant recycling and disposal laws.

Note: Use and store batteries in a dry environment.

Caution: Do not charge the battery under direct sunlight.

Caution: Do not charge the battery when the battery feels hot.

Caution: Always be present when charging batteries and **do not leave batteries connected to the chargers or the QDrone overnight**.

Caution: Charge and store LiPo batteries in a location where a battery fire or explosion (including smoke hazard) will not endanger life or property.

Caution: Keep LiPo batteries away from children and animals.

Caution: Never charge a LiPo battery that has ballooned or swollen due to overcharging, undercharging or from a crash.

Caution: Never charge a LiPo battery that has been punctured or damaged in a crash. After a crash, inspect the battery pack for signs of damage.

Caution: Never charge the LiPo battery in a moving vehicle.

Caution: Never overcharge the LiPo battery.

Caution: Never leave the LiPo battery unattended during recharging.

Caution: Do not charge LiPo batteries near flammable materials, liquids or objects.

Caution: Ensure that charging leads are connected correctly. Reversing polarity charging can lead to battery damage, fire or explosion.

Caution: A LiPo battery fire is a chemical fire. Have a suitable fire extinguisher (class D/for electrical fires) or a large bucket of dry sand near the charging area. Do not try to extinguish electrical battery fires with water.

Caution: Reduce risks from fire/explosion by storing and charging LiPo batteries inside a suitable container: a LiPo storage sack/bag or metal/ceramic container is advised.

Caution: Protect your LiPo batteries from accidental damage during storage and transportation. Do not put battery packs in pockets or bags where they can short circuit or come into contact with sharp or metallic objects.

Caution: If your LiPo battery is subjected to a shock (such as a crash) you should place it in a metal container and observe signs of swelling or heating for at least 30 minutes.

Ensure that the metal container doesn't short the leads of the battery, which may cause a fire!

Caution: Do NOT attempt to disassemble, modify, or repair the LiPo battery.

Caution: Never use a battery that is warm from charging or charge a battery that is warm from usage.

Note: Consider how you would deal with a LiPo battery fire/explosion as part of your normal fire safety and evacuation planning.

Note: When discarding a LiPo battery, discard it in accordance with your country's relevant recycling and disposal laws.

Note: Monitor charging LiPo batteries for signs of overheating.

B. QDrone 1 Batteries

QDrone 1 is powered using a single pack Lithium-Polymer (LiPo) 3S 3300mAh battery (Figure 1a). For safety, these batteries are shipped uncharged and must be charged prior to first use. Check the voltage of the supplied batteries using the RC battery voltage tester provided (Figure 1b). Note that the black cable (ground) goes to the first pin. For the supplied batteries, the safe and unloaded operating ranges are between 10.0V and 12.6V, with the latter being the maximum voltage. If the battery's voltage is below 12.4V, it must be charged prior to use with the QDrone 1.

a. LiPo 3s 3300mAh with balancer cable (4 wire connector) and XT60 female connector (2 wire connector)

b. RC battery voltage tester

Figure 1: Battery voltage testing and connection in QDrone 1

Caution: A battery voltage below 10V increases the risk of uneven charge between the three cells. If you experience issues charging a battery that is consistently below 10V, discard it in accordance with your country's relevant recycling and disposal laws.

C. Charging the QDrone 1 Batteries

1. Power the battery charger/balancer (Figure 2a) with the supplied power supply (Figure 2b).
2. Connect the 4 pin battery balancer cable to the 4 pin connector for the battery charger (Figure 2c).
3. The top LEDs on the EV-Peak charger will indicate the status of the battery from low to full charge.

a. charger / balancer | b. charger/balancer power supply cable | c. QDrone 1 connection to charger

Figure 2: Wiring and using the RC battery charger/balancer

Note: When the charger/balancer beeps continuously, or the status on the charger stopped flashing at 100%, the battery is charged.

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